



BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, Pilani
Pilani Campus
AUGS/ AGSR Division

First Semester 2026-27
COURSE HANDOUT (Part II)

Date: 02.08.2026

In addition to Part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : EE G501
Course Title : Environmental Sampling and Analytical Methods
Instructor-in-Charge : PUBALI MANDAL

1. Course Description: Principles of sample collection and data analysis/interpretation, Gravimetric methods, titrimetric methods, electrochemical methods, Spectrometric method of analysis, Chromatographic method of analysis, sampling techniques for air and water pollution, Biological methods of analysis, Interpretation of data in environmental monitoring.

2. Scope and Objective of the Course: The course introduces methodology of sample collection (sampling techniques) for water/wastewater/air. The course covers various aspects of sample analysis through conventional (for example gravimetric, titrimetric) and advanced methods (for example electrochemical, spectrophotometric, chromatographic). The course will also guide students, to apply concepts for environmental monitoring.

3. Text Books:

T1. Zhang, C., 2007. Fundamentals of environmental sampling and analysis. John Wiley & Sons.

T2. Csuros, M., 2018. Environmental sampling and analysis: lab manual. Routledge.

4. Reference Books/Journals:

R1. Csuros, M. and Csuros, C., 2016. Environmental sampling and analysis for metals. CRC Press.

R2. Popek, E.P., 2017. Sampling and analysis of environmental chemical pollutants: a complete guide. Elsevier.

R3. Rice, E.W., Bridgewater, L. and American Public Health Association eds., 2012. Standard methods for the examination of water and wastewater (Vol. 10). Washington, DC: American public health association.

R4. Rong, Y. ed., 2011. Practical environmental statistics and data analysis. ILM publications.

R5. Ott, W.R., 1994. Environmental statistics and data analysis. CRC Press.

R6. Journals: Elsevier, Springer, Taylor & Francis etc.

5. Course Plan:



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Lecture no.	Learning objectives	Reference	Topics to be covered
1-5	Basics of Environmental Sampling and Analysis	T1	Essential analytics of environmental sampling, environmental statistics, environmental regulations
6-9	Environmental Sampling Techniques	T1/R2	General Guidelines of Environmental Sampling Techniques, Techniques for Sampling in various Media (air and water)
10-14	Gravimetric methods	T1/Journals/ T2	Precipitation Gravimetry, Volatilization Gravimetry, Particulate Gravimetry
15-18	Titrimetric methods	T1/T2	Acid-Base titration
19-21	Electrochemical methods	T1	Potentiometric and Voltammetric Applications
22-26	Spectrometric methods of analysis	T1/R1	UV-Visible and Infrared Spectroscopic Methods in Environmental Analysis, Atomic Spectroscopy
27-32	Chromatographic methods of analysis	T1	Gas Chromatography, High Performance Liquid Chromatography, Ion Chromatography
33-35	Biological methods of analysis	R6	Principle of biological methods, indicator organisms, concentration and extraction
36-38	Interpretation of data in Environmental monitoring	R4/R5	Quality Assurance/Quality Control, Statistical application in environmental monitoring
39-40	Demonstration of analysis methods	Lecture notes	Spectrometric, High Performance Liquid Chromatography

6. Evaluation Scheme:

Component	Duration	Weightage (%)	Date & Time	Nature of component
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				(Close Book/ Open Book)
Mid-Semester Test	90 Min	30	8/10 FN2 (see timetable)	CB/OB
Comprehensive Examination	3 h	40	1/12 AN (see timetable)	CB/OB
3 Class tests (Best 2)	Throughout the Semester	15	To be announced in the class	CB/OB
Project seminar/Report	Throughout the Semester	15	To be announced in the class	CB/OB

CB: Close Book, OB: Open Book

7. Chamber Consultation Hour: Thursday: 5-6 PM

8. Notices: Watch *BITS e-Mail/Google classroom/Nalanda*

9. Make-up Policy:

1. Make-up will be granted only on genuine reasons.
2. For medical cases, a certificate from the Chief Medical Officer of the Medical Centre must be produced.
3. Make up will not be conducted for class tests. However, best of two will be considered for final evaluation.

10. Grading Policy:

If a student absents himself/herself in any component (listed in the Evaluation Scheme), his/her performance in the course may be reported as 'NC' (Not Cleared). A student should obtain 30% of the average of the top few performers in the class, or 40% of the median marks of the class, whichever is lower, to clear the course.

Note (if any):

- Students have to adhere to the class timings and participate in classroom discussions.
- Academic honesty and academic integrity are to be maintained by all of the students throughout the Semester.

PUBALI MANDAL
Instructor-in-charge



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